

ProGear Ruggedized Portable and 5000 ProGear Series Pumps



Operational and Installation Guidelines

Please read these Operational and Installation Guidelines before installing the ProFlo™ Series Pumps. If additional help is needed, please consult the Factory.

CAUTIONS

- Water pipe connections and fixtures directly connected to a potable water supply shall be sized, installed and maintained in accordance with Federal, State, and local codes.
- Do not operate the pump above the pressure limitations specified on the data label.
- Never operate the pump in a harsh environment or hazardous atmosphere, since motor brush and switch may cause electrical arcing.
- **Do not pump gasoline or other flammable liquids. Do not use with petroleum products.**
- **Do not assume fluid compatibility. If the fluid is improperly matched to the pumps' elastomers, a leak may occur**
- As long as there is inlet water pressure, the pump will not stop forward flow of water even if the motor is turned off. Be sure the system has positive means of shutting off water supply.
- Always consider electrical shock hazard when working with and handling electrical equipment. If uncertain, consult an Electrician. Electrical wiring should only be done by a qualified Electrician per Local and State Electrical Codes.

MOUNTING

- The pump should be mounted in a dry place and away from any source of heat. If an enclosure is used, special instructions for cooling the motor may be necessary. Consult the Factory.
- Do not subject the pump to extreme high or low (freezing) temperatures while in operation. (Operating ambient temperature range is 32°F to 115°F).

Installation Procedure

The basic demand pump is controlled by a built-in pressure sensing demand switch. When a faucet or valve is opened down stream of the pump, line pressure drops, thus starting the pump automatically. Conversely, when the valve shuts, the line pressure increases turning the pump off automatically. The pressure switch actuates in response to the pump outlet pressure at a predetermined and preset pressure. The pump label indicates the pre-set OFF pressures. Typically, the OFF pressure is accurately set at the FACTORY and the ON pressure is within an allowable range below that value. In response to the characteristics of the system in which the pump is installed, such as the flexibility and length of the tubing, and the faucet or valves and the duration that they are open, these pressure settings may vary. Therefore, change in pressure settings is expected with use and over time. If the pump does not have an integral pressure sensing demand switch (i.e. pump is operated with an external control), pump will be equipped with a bypass relief valve (bypass is factory preset).

Read the OPERATIONAL AND INSTALLATION GUIDELINES on the other side carefully before starting to install the pump. Consult the Factory if there is any question.

Determine the optimum location for your pump before proceeding.

- Turn off the fluid supply.
- Cut the flexible tubing in sufficient length to avoid any stress on the tubing or hose where it connects to the pump or the fitting on any accessory or hose.
- Insert tubing or hose into pump ports. If compression fittings with threaded nuts are used, insert tubing until it bottoms out in the port and hand tighten the compression nut until the connection is tight. Then use a wrench to tighten the

- The pump may be mounted horizontally with the outlet port on the right when viewed from the pump end or with the pump above the mount; or vertically with the pump above or below the motor.

PLUMBING

- The pump is equipped with either a pressure sensing demand switch, a bypass relief valve, or both which controls the maximum safe operating pressure.
- We recommend use of flexible tubing with proper pressure rating.
- Pump will prime only if all pressure is relieved from outlet port.
- It is recommended that debris-free liquid be pumped or an in-line sediment filter (50 mesh) be installed at the inlet side to keep foreign debris out of the system.
- The pump should always be mounted prior to any components which could introduce particles to the liquid; thus, preventing them from entering the pump chambers and possibly causing clogging.
- Avoid any sharp bends which may crimp tubing and restrict flow. Use 90° elbow fittings if necessary. REMCO provides pumps with different kinds of fittings. Please consult Factory for your needs.

ELECTRICAL

- The ProFlo™ Series Pumps are designed for intermittent duty, but may run continuously if the motor temperature does not exceed the recommended limit. Some ProFlo™ Series Pumps are equipped with thermally protected motors and in case motor temperature exceeds thermal cut-out rating, pump will shut down and will not restart until motor cools down to specified temperature. Please consult the appropriate Data Sheet for continuous running parameters, noting the effect of rapid On/Off Cycling (See section below).
- If a power supply is used with the system and the supply is not furnished by REMCO, it will need to be reviewed for correct application and approval by REMCO.

nut 1/2 turn clockwise or follow the wrench tightening instructions provided by the fitting manufacturer.

- The ProFlo™ Series pump is now ready for operation. Open the inlet valve if any to allow fluid to flow to the pump.
- If the power source is a transformer, plug the appropriate REMCO supplied or approved transformer into the receptacle and connect the pump to the transformer. If the power source is not a transformer, connect the pump to the appropriate power source. Open the discharge or dispensing valve. Allow water to circulate, purging any entrapped air.
- The pump will now start building pressure. Operating pressure will vary with flow rate, flow valve, feed-water pressure and line voltage. Check for fitting leaks.
- If compression fittings with threaded nuts are used, observe any leaks after pump has run for approximately 15 minutes. Further tighten compression nuts approximately 1/8 to 1/4 of a turn on all fittings in the system. Wait 15 minutes and repeat the leak check.

NOTE: Further adjustments should not be necessary although it may take several days of operation before all the air has been purged and the system is stabilized.

- **Rapid On/Off Cycling must be limited to no more than 6 times per minute, even if the pump is operating in the Continuous Duty zone. Cycling could cause the motor to heat beyond the recommended maximum temperature, and reduce the operational life of the pump and pressure-sensing switch.**

Servicing

EVERY USE

Run clean water through the pump and system.

EVERY YEAR

Check system against operating standards. If continuous duty, replace lower housing assembly

EVERY OTHER YEAR

Check against operating standards. If continuous duty, replace motor

EVERY FIFTH YEAR

Replace valves

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Troubleshooting Guidelines

IF PUMP WILL NOT START CHECK:

- Correct voltage ($\pm 10\%$) and electrical connections
- Fuse or breaker
- Pressure switch operation and correct voltage at switch
- Rectifier or motor for open or grounded circuit

IF PUMP WILL NOT PRIME (NO DISCHARGE WITH MOTOR RUNNING) CHECK:

- Debris in strainer
- Restriction (kinks) in inlet/outlet tubes
- Debris in inlet/outlet
- Air leaking in from inlet

IF PUMP WILL NOT SHUT OFF (OUTPUT LINE CLOSED AND NO LEADS) CHECK:

- Air trapped in outlet line or pump head
- Correct voltage to pump
- Debris in pump inlet/outlet valves
- Loose drive assembly or pump head screws
- Pressure switch operations/adjustments
- Leaks

IF PUMP LEAKS FROM PUMP HEAD OR SWITCH CHECK:

- Loose screws at switch or pump head
- Switch diaphragm ruptured or pinched
- Punctured diaphragm if fluid is present

Warranty

No Hassle Warranty- ProGear 5000 Series Water Pumps and ToughGear Portable Pumps

We will replace or exchange this product at no or limited cost within 2 years of its original purchase based on the following conditions:

USA Adventure Gear warrants all pump series to be free from material and workmanship defects (under normal use and service) for a period of two (2) years from the date of manufacture. This warranty does not extend to any products which have been altered or modified; to any products that have not been properly maintained; nor to units that are returned in an unassembled condition. The warranty guarantees that products will perform to USA Adventure Gear's flow and pressure specifications throughout the life of the warranty. The warranty does not cover wear, appearance, misuse, abuse, misapplication, or external water damage. If the returned product is found not to be defective under the conditions of this warranty, a charge may be made for repair or replacement.

This is a Limited Warranty. It covers the product only and the extent of the coverage is limited to the cost of the product. As the manufacturer has no control over shipping, handling and installation, the warranty cannot cover water damage, or any other damage, caused by a leak or other malfunction.

THIS WARRANTY IS IN LIEU OF ALL WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING

IMPLIED WARRANTIES OR WARRANTIES OF MERCHANTABILITY AND/OR FITNESS FOR A

PARTICULAR PURPOSE. NO PERSON IS AUTHORIZED TO GIVE ANY OTHER WARRANTY OR

ASSUME OBLIGATION OR LIABILITY ON USA ADVENTURE GEAR'S BEHALF. USA ADVENTURE GEAR'S SHALL NOT BE LIABLE FOR ANY INDIRECT, INCIDENTAL, PUNITIVE OR CONSEQUENTIAL DAMAGES OF ANY KIND INCURRED BY THE REASON OF THE USE OR SALE OF ANY DEFECTIVE PRODUCT AND PART.

THIS WARRANTY SHALL BE GOVERNED BY THE LAWS OF THE STATE OF COLORADO, WITH

ANY DISPUTE OR CLAIM VENUED OR HEARD IN THE STATE OF COLORADO.



WARNING: this product can expose you to chemicals including "carbon black", which is known to the state of California to cause cancer. For more information go to www.P65Warnings.ca.gov

Replacement Parts

1 Lower Housing Assembly

PART NUMBER	GPM	DESCRIPTION
LHA5513-1E77	5.3GPM	DEMAND OR BYPASS, SANTOPRENE

2 Valve Housing Assembly

PART NUMBER	GPM	DESCRIPTION
VHA5513-1E77	3.2-5.3GPM	DEMAND OR BYPASS, VITON VALVES

3 Upper Housing Assembly

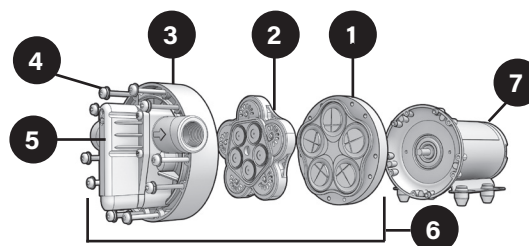
PART NUMBER	GPM	DESCRIPTION
UHA5503-1E77	3.2-5.3GPM	3/4" QUICK ATTACH W/PRESSURE SWITCH

4 Mounting Bolt Kit

PART NUMBER	GPM	DESCRIPTION
55FB-SK	3.2-5.3GPM	10 MOUNTING BOLTS W/WASHERS

5 Pressure Switch Assembly

PART NUMBER	GPM	DESCRIPTION
PSW-580R-60	3.2-5.3GPM	60PSI PRESSURE SWITCH



6 Complete Pumphead Assembly

PART NUMBER	GPM	DESCRIPTION
5503-1X77	5.3GPM	3/4" QUICK ATTACH W/PRESSURE

7 Motor

PART NUMBER	GPM	DESCRIPTION
M582-S	3.2-5.3GPM	SEALED DEMAND MOTOR W/2 PIN CONNECTOR
M70-563-JDS	3.2-5.3GPM	SEALED BYPASS MOTOR W/2 PIN CONNECTOR